Unselective Binder of Wh-dependency under Optimality Theory

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Empirical Data

Wh-island & Complex-NP island in Chinese English and Japanese

English

- English gives rise to both wh-island effects and complex-NP island effects (Ross 1967, Chomsky 1977 et.al.):

Wh-island: *Who$_j$ do [$_{TP}$ you think [$_{CP}$ why$_i$ [$_{TP}$ John met t_i t_j]]]?

Complex-NP: *Who$_j$ do you like [$_{DP}$ the books$_k$ [$_{CP}$ Op$_k$ that [$_{TP}$ t_j wrote t_k]]]?

Chinese

- Chinese does not give rise to any of these island effects (for wh-nominals) (Huang 1982, Cheng 1991 et.al.):

Wh-island:

[$_{CP}$ [$_{TP}$ John xiang zhidao [$_{CP}$ [$_{TP}$ shei qu-le nali]]]]?
John wonder who go-Perf where

(i) ‘For which person x, such that John wonders where x went?’
(ii) ‘For which place y, such that John wonders who went to y?’

Complex-NP:

[$_{CP}$ [$_{TP}$ Ni xihuan [$_{DP}$ [$_{CP}$ Op$_k$ [$_{TP}$ shei xie] de] shu$_k$]]]?
2SG like who write C book

‘For which person x, such that you like the books that x wrote?’

Japanese

- Japanese only gives rise to wh-island effects but not to complex-NP island effects (Nishigauchi 1986, Watanabe 1992 et.al.):

Wh-island:

* [$_{CP1}$ [$_{C'}$ [$_{TP1}$ John-wa [$_{CP2}$ [$_{TP2}$ Mary-ga [$_{DP}$ [$_{NP}$ nani]]-o katta]
John-Top Mary-Nom what-Acc buy
ka-dooka] Tom-ni tazuneta] no]]?
whether Tom-Dat ask Q
'What x, such that John asked Tom if Mary bought x?'

Complex-NP:

[$_{CP1}$ [$_{C'}$ [$_{TP1}$ John-wa [$_{DP}$ [$_{NP}$ [$_{CP2}$ Op$_k$ [$_{TP2}$ t$_k$ dare-o aisiteiru]]
John-Top who-Acc love
onna$_k$]]-o nagutta] no]]?
woman-Acc beat Q
'For which person x such that John beat the woman [who loves x]?'

Summary of the differences:

	Chinese	Japanese	English
Wh-island	×	√	√
Complex-NP	×	×	√

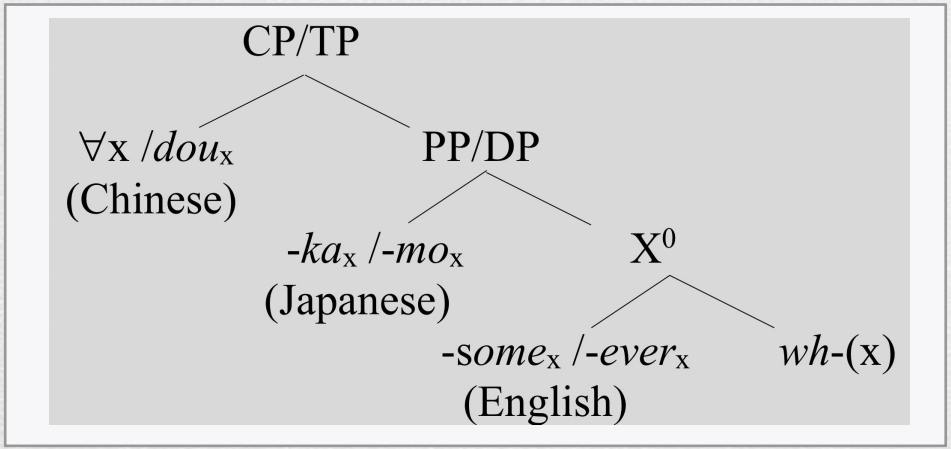


Building Block 1

Unselective Binding (Tsai 1994)

Tsai (1994): Unselective Binding

- Tsai assumes that wh-dependencies in all languages are formed by a null Op operator and an in-situ nominal restriction of wh-words. Op can be assumed to be a part of the lexical item (i.e., wh-words).
- Op is generated in different positions in English, Japanese and Chinese, and Op is moved to [Spec, CP] at surface structure. Since such a movement is an A bar movement at the level of overt syntax (i.e., surface structure), it obeys locality constraints.



English: Word Level Op

- English Q-operator is generated below X^0 level, and the whole wh-phrase must be moved to the Spec of CP: both Complex NP and wh-island effects are observed: $[_{CP}\ [_{DP}\ \textit{wh}(x)\text{-}Op_{x[Q]}]_j \quad [_{TP}\ \ldots\ t_j\ \ldots\ ]]$



- The movement of Op will force the morphological form of the entire wh-word to move together and such a movement will necessarily cross both complex-NP island and wh-island:

  Wh-island: $*[_{CP}\ [_{DP}\ \text{who-Op}_{[Q]}]_j\ [_{TP}\ \ldots\ [_{CP}\ [_{AP}\ \text{why-Op}]_i\ [_{TP}\ \ldots\ t_i\ t_j]]]]$

  Complex-NP: $*[_{CP}\ [_{DP}\ \text{who-Op}_{[Q]}]_j\ [_{TP}\ \ldots\ [_{NP}\ \text{books}_k\ [_{CP}\ Op_k\ [_{TP}\ t_j]]]]]$

Chinese: Sentential Level Op

- In Chinese, a null interrogative operator is generated at the sentential level (i.e., TP/CP level). The Op unselectively binds the in-situ wh-word as variable. Since no movement is involved and no island is crossed, no locality effects are observed: $[_{CP}\ Op_{x[Q]}\ [_{TP}\ \ldots[_{DP}\ \textit{wh}(x)\ \ldots\ ]]$

Wh-island: $[_{CP}\ Op_{j[Q]}\ [_{TP}\ \ldots\ [_{CP}\ Op_i\ [_{TP}\ [_{DP}\ [_{NP}\ shei_j]]]\ldots[_{AP}\ nali_i]]]]]$
who where

Complex-NP: $[_{CP}\ Op_{j[Q]}\ [_{TP}\ \ldots\ [_{NP}\ [_{CP}\ Op_k\ [_{TP}\ [_{DP}\ [_{NP}\ shei_j]]]\ldots shu_k]]]]]$
who book

Japanese: Phrasal Level Op

- Japanese Q-operator is located at the Spec of DP, the 'halfway' movement to the Spec of CP avoids Complex NP island, but still exhibits wh-island effects: $[_{CP}\ Op_{x[Q]}\ [_{TP}\ \ldots\ [_{DP}\ t_x\ [\ \ldots\ wh(x)\ \ldots\ ]]]]$



- Q Op is higher than Complex-NP island. The movement of Q will not cross Complex-NP island, therefore Japanese lacks Complex NP effects:
  Complex-NP: $[_{CP}\ Op_{j[Q]}\ [_{TP}\ [_{DP}\ t_j\ [_{NP}[_{CP}\ Op_k\ [_{TP}\ [_{DP}\ [_{NP}\ dare_j]]\text{-}o]]\ onna_k]]\text{-}o\ldots]]?$

- In wh-island context, DP is still below the intermediate CP (= wh-island). The movement of Q will necessarily cross the wh-island, which gives rise to wh-island effects:
  Wh-island: $*[_{CP}\ Op_{j[Q]}\ [_{TP}\ldots[_{CP}\ Op_i\ [_{TP}\ldots[_{DP}\ t_j\ [_{NP}\ nani_j]]\text{-}o\ldots[_{AP}\ ka\text{-}dooka_i]]]]]$



Building Block 2

Optimality Theory in Syntax

What is Optimization

"... a grammatical procedure which chooses among different potential alternatives and blocks sub-optimal alternatives, based on a structured set of criteria..."

(Trommer 2020)

		CoolPeople	CoolData	JobOpportunity
a.	Morphology	*!		**
b. ☞	Phonology		*	***
c.	Syntax	*!**	**	*
d.	Semantics	*!	***	*

Minimal Link Condition (MLC)

No movement that crosses another variable binder (Op)!

- Minimal Link Condition (MLC): If in a structure Σ = H...[...α...[...β...]...] (where H c-commands α and β, and α asymmetrically c-commands β) both α and β are of the right type to establish a relation R with H, then H can establish R only with α (but not with β) (Chomsky 1995).

English		MLC
Wh-island: *[$_{CP}$ **Who** do [$_{TP}$ you think [$_{CP}$ why [$_{TP}$ John met _]]]]?	*[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ ... [$_{CP}$ [$_{AP}$ why-Op] [$_{TP}$ ... **t$_j$**]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ ... [$_{CP}$ Op$_i$ [$_{TP}$ ...[$_{AP}$ why$_i$][$_{DP}$ **t$_j$** [$_{NP}$ **who$_j$**]]]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ ... [$_{CP}$ Op$_i$ [$_{TP}$ ...[$_{AP}$why$_i$][$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]	
Complex NP island: *[$_{CP}$ **Who** do [$_{TP}$ you like [$_{DP}$ the books [$_{CP}$ that [$_{TP}$ _ wrote]]]]]?	*[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ ... [$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ **t$_j$**]]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]]]	
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ ...[$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ ...[$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]]	

Minimal Binding (MB)

No bounding nodes between the Op and the wh-words it binds!

- Following Chomsky 1986, NPs and TPs are considered as bounding nodes:

English		MB
No island: [$_{CP}$ **Who** do [$_{TP}$ you think [$_{CP}$ that [$_{TP}$ John met _]]]]?	[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ t_j' that [$_{TP}$ … t_j]]]]	
	*[$_{CP}$ Op$_{j[Q]}$ [$_{TP}$ … [$_{CP}$ t_j' that [$_{TP}$ …[$_{DP}$ **t_j** [$_{NP}$ **who$_j$**]]]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ [$_{TP}$ …[$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]	*!**

Economy (E)

No movement! No trace!

- Each step of the movement (trace) would be counted as one time violation:

English		E
No island: [$_{CP}$ **Who** do [$_{TP}$ you think [$_{CP}$ that [$_{TP}$ John met _]]]]?	[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ **t$_j$'** that [$_{TP}$ … **t$_j$**]]]]	*!*
	*[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ that [$_{TP}$ … **t$_j$**]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ **t$_j$'** that [$_{TP}$ …[$_{DP}$ **t$_j$** [$_{NP}$ who$_j$]]]]]]	*!*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ that [$_{TP}$ …[$_{DP}$ **t$_j$** [$_{NP}$ who$_j$]]]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ [$_{TP}$ …[$_{DP}$ [$_{NP}$ who$_j$]]]]]]	

Economy (E)

Phase theory (Chomsky 2008) is not something we can apply whenever we want! -> CP only has one Specifier!

- The successive-cyclic wh-movement is blocked either by the second wh-variable's operator, or the operator co-indexed with the relativized noun, both of which are generated in the intermediate [Spec CP]:

Chinese		E
Wh-island: [$_{CP}$ [$_{TP}$ John xiang zhidao [$_{CP}$ [$_{TP}$ **shei** qu-le nali]]]]?	*[$_{CP}$ [$_{DP}$ **shei-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ [$_{AP}$ nali-Op]…[$_{TP}$ … **t$_j$**]]]]	*!
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$ **shei$_j$**]]…[$_{AP}$ nali$_i$]]]]]	*!
	[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **shei$_j$**]]…[$_{AP}$ nali$_i$]]]]]	



Implementation

Different languages have different rankings of constraints.

English: Word Level Op

English		MB	MLC	E
No island: [$_{CP}$ **Who** do [$_{TP}$ you think [$_{CP}$ that [$_{TP}$ John met _]]]]?	[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ **t$_j$**’ [$_{TP}$ … **t$_j$**]]]]			**
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ **t$_j$**’ [$_{TP}$ …[$_{DP}$ **t$_j$** [$_{NP}$ **who$_j$**]]]]]]	*!		**
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ [$_{TP}$ …[$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]	*!**		
Wh-island: *[$_{CP}$ **Who** do [$_{TP}$ you think [$_{CP}$ why [$_{TP}$ John met _]]]]?	*[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ [$_{AP}$ why-Op] [$_{TP}$ … **t$_j$**]]]]		*!	*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ …[$_{AP}$ why$_i$][$_{DP}$ **t$_j$** [$_{NP}$ **who$_j$**]]]]]]	*!	*	*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ … [$_{AP}$why$_i$][$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]	*!**		
Complex NP island: *[$_{CP}$ **Who** do [$_{TP}$ you like [$_{DP}$ the books [$_{CP}$ that [$_{TP}$ _ wrote]]]]]?	*[$_{CP}$ [$_{DP}$ **who-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ **t$_j$**]]]]]		*!	*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]]	*!**		*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ …[$_{NP}$ books$_k$ [$_{CP}$ Op$_k$ [$_{TP}$ …[$_{DP}$ [$_{NP}$ **who$_j$**]]]]]]	*!***		

Chinese: Sentential Level Op

Chinese		E	MLC	MB
Wh-island: [$_{CP}$ [$_{TP}$ John xiang zhidao [$_{CP}$ [$_{TP}$ **shei** qu-le nali]]]]?	*[$_{CP}$ [$_{DP}$ **shei-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{CP}$ [$_{AP}$ nali-Op]…[$_{TP}$ … **t$_j$**]]]]	*!	*	
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$ **shei$_j$**]]…[$_{AP}$ nali$_i$]]]]]	*!	*	*
	[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{CP}$ Op$_i$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **shei$_j$**]]…[$_{AP}$ nali$_i$]]]]]			***
Complex NP island: [$_{CP}$ [$_{TP}$ Ni xihuan [$_{NP}$ [$_{CP}$ [$_{TP}$ **shei** xie] de] shu]]]?	*[$_{CP}$ [$_{DP}$ **shei-Op$_{[Q]}$**]$_j$ [$_{TP}$ … [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ **t$_j$**]] shu$_k$]]]	*!	*	
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **shei$_j$**]]…shu$_k$]]]]]]	*!		***
	[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ … [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **shei$_j$**]]…shu$_k$]]]]]			****

Japanese: Phrasal Level Op

Japanese with wh-island effect (Nishigauchi 1986, Watanabe 1992 et.al.)		MLC	MB	E
Wh-island: *[$_{CP}$ [$_{TP}$ John-wa [$_{CP}$ [$_{TP}$ Mary-ga [$_{DP}$ [$_{NP}$ **nani**]]-o katta] ka-dooka] Tom-ni tazuneta] no]?	*[$_{CP}$ [[$_{DP}$ **nani-Op$_{[Q]}$**]-o]$_j$ [$_{TP}$…[$_{CP}$ [$_{AP}$ ka-dooka-Op] [$_{TP}$ **t$_j$**]]]]	*!		*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$…[$_{CP}$ Op$_i$ [$_{TP}$…[$_{DP}$ **t$_j$** [$_{NP}$ **nani$_j$**]]-o…[$_{AP}$ ka-dooka$_i$]]]]]	*!	*	*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$…[$_{CP}$ Op$_i$ [$_{TP}$…[$_{DP}$ [$_{NP}$ **nani$_j$**]] -o…[$_{AP}$ ka-dooka$_i$]]]]]		***	
Complex NP island: [$_{CP}$ [$_{TP}$ John-wa [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ t$_k$ **darei-o** aisiteiru]] onna$_k$]-o nagutta] no]?	*[$_{CP}$ [[$_{DP}$ **dare-Op$_{[Q]}$**]-o]$_j$ [$_{TP}$…[$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ **t$_j$**]] onna$_k$]…]]?	*!		*
	[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$ [$_{DP}$ **t$_j$** [$_{NP}$[$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **dare$_j$**]]-o]] onna$_k$]]-o…]]?		***	*
	*[$_{CP}$ **Op$_{j[Q]}$** [$_{TP}$…[$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **dare$_j$**]]-o…]] onna$_k$]…]]?		*!***	



Challenge

Does Japanese have true Wh-island?

Japanese: Sentential Level Op

Japanese without wh-island effect (Lasnik & Saito 1984,1992 et.al.)		E	MLC	MB
Wh-island: [$_{CP}$ [$_{TP}$ John-wa [$_{CP}$ [$_{TP}$ Mary-ga [$_{DP}$ [$_{NP}$ **nani**]]-o katta] ka-dooka] Tom-ni tazuneta] no]?	*[$_{CP}$ [[$_{DP}$ **nani-Op**$_{[Q]}$]-o]$_j$ [$_{TP}$…[$_{CP}$ [$_{AP}$ ka-dooka-Op] [$_{TP}$ **t**$_j$]]]]	*!	*	
	*[$_{CP}$ **Op**$_{j[Q]}$ [$_{TP}$…[$_{CP}$ Op$_i$ [$_{TP}$…[$_{DP}$ **t**$_j$ [$_{NP}$ **nani**$_j$]]-o…[$_{AP}$ ka-dooka$_i$]]]]]	*!	*	*
	[$_{CP}$ **Op**$_{j[Q]}$ [$_{TP}$…[$_{CP}$ Op$_i$ [$_{TP}$…[$_{DP}$ [$_{NP}$ **nani**$_j$]] -o…[$_{AP}$ ka-dooka$_i$]]]]]			***
Complex NP island: [$_{CP}$ [$_{TP}$ John-wa [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ t$_k$ **darei-o** aisiteiru]] onna$_k$]-o nagutta] no]?	*[$_{CP}$ [[$_{DP}$ **dare-Op**$_{[Q]}$]-o]$_j$ [$_{TP}$…[$_{NP}$[$_{CP}$ Op$_k$ [$_{TP}$ **t**$_j$]] onna$_k$]…]]?	*!	*	
	*[$_{CP}$ **Op**$_{j[Q]}$ [$_{TP}$ [$_{DP}$ **t**$_j$ [$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **dare**$_j$]]-o]] onna$_k$]]-o…]]?	*!		***
	[$_{CP}$ **Op**$_{j[Q]}$ [$_{TP}$…[$_{NP}$ [$_{CP}$ Op$_k$ [$_{TP}$ [$_{DP}$ [$_{NP}$ **dare**$_j$]]-o…]] onna$_k$]…]]?			****

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Thank You !

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